# SIMATS ENGINEERING

## ASSESSMENT TOOL 1: Analytical Problem Solving

**COURSE NAME:** Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

**COURSE OUTCOME COVERED**

**CO2:** *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 50**

Question Count: 5

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**Code of Conduct** C192525122)

I K.Maheshbabu (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student. K.Maheshbabu.

**Assessment Tasks**

**Analytical Problem Solving**

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

| Criteria | Weightage | Level 4 - Exemplary | Level 3 – Proficient | Level 2 - Developing | Level 1 - Beginning |
|---|---|---|---|---|---|
| Problem Interpretation | 20% | Interprets all technical requirements accurately | Interprets most requirements correctly | Partial interpretation | Misinterprets the problem |
| Analytical Reasoning | 25% | Strong reasoning with clear cause-effect analysis | Good reasoning with minor gaps | Basic analysis only | Weak or no analysis |
| Method Selection | 20% | Chooses highly appropriate virtualization and security concepts | Chooses mostly suitable concepts | Partially suitable concepts | Inappropriate approach |
| Solution Quality | 20% | Provides feasible and technically sound solution | Reasonable solution with minor issues | Basic solution with limitations | Unclear or invalid solution |
| Presentation of Analysis | 15% | Well-structured and technically precise | Generally clear | Somewhat unclear | Difficult to follow |

**Total Marks Awarded:**

Compare Type-1 and Type-2 hypervisors for a Cloud data Center hosting mission-critical application. Analyze Performance, Security and manageability, then justify the better choice.

A. Introduction:-

Virtualization is a technology that allows multiple Virtual machines (VMs) to run on a Single physical Server. The Software responsible for Creating and managing these VMs are called a "Hypervisor. Hyper-Visors are classified into two categories. Type(1) that is (Bare-Metal) and Type-2 (Hosted) hypervisors.

Type-1 Hypervisor:-

A Type-1 hypervisor runs directly on the physical hardware without requiring a host operating system. It acts as an interface between the hardware and virtual Machiness.

Ex:-

- VMware ESXi
- Microsoft Hyper-V
- Xen Server.

Because, it directly accesses hardware resources, it provides better Performance, improved Security, and Greater realibility.

## Type-2 Hypervisor:-

A Type-2 hypervisor runs top of a host operating System such as windows (or) Linux. Virtual Machines operate through the hypervisor, which relies on the host OS for hardware access.

Ex:-

- Oracle Virtual Box.
- VMware Workstation.
- Parallels Desktop.

Type-2 hypervisors are mainly used for testing, development, and Educational Purposes rather than Enterprise cloud Environments.

## Security analysis:-

Security is essential for Protecting Sensitive of Organizational Data.

Type-1 Hypervisor have a smaller attack Surface because there is no host operating System. fewer Software layers reduce Security Vulnerabilities & unauthorized access risses.

Type-2 Hypervisors depend on the host operating System. if the most os becomes infected by malware (or) Compromised by attackers, all Virtual machines may also be affected.

Thus, Type-1 and hypervisors Provide stronger Security and better Workload isolation.

A Cloud Provider must isolate workloads belonging to finance, healthcare, and Student Portals on the Same hardware. Analyze how Virtualization of architecture Supports isolation and identify two major attacks Surfaces.

A. Introduction:

Cloud Computing allows multiple application and organizations to share the same physical infrastructure. for Examples a Cloud Provider may host finance Systems, health care applications & Student portals on a Single Server.

Virtualization Architecture:

1. physical Hardware 2. Hypervisor 3. Virtual Machine
4. Virtual Network infrastructure 5. Virtual storage resources

How Virtualization Supports isolation:

1. CPU isolation:

Each VM gets its own CPU resources preventing interference from other VMs.

2. Memory isolation:

One VM Cannot access the RAM of another VM.

3. Storage isolation:

Each VM has its own Virtual disk and data storage

4. Network isolation:

Virtual networks and applications virtual Switches Keep VM traffic Separate.

5. Access Control:-
users and applications can only access their assigned VM resources.

Benefits of Virtualization - Based isolation:-

- Improved Security
- Better resource utilization.
- Reduced hardware Costs.
- Independent workload management.
- Easier backup and recovery.

Major attack Surface 1:- Hypervisor layer:-
Hypervisor is the most Critical Component in a Virtualized Environment because it controls all Virtual machines.

Threats:-

- VM Escape Attacks.
- Hypervisor Exploits.
- Privilege Escalation.

Major attack Surface 2:- Virtual Network infrastructure:-
Virtual switches, Virtual routers, and Software-defined works networks Create another important attack Surface.

Threats:-

- Packet Sniffing
- ARP spoofing
- Network Scanning
- Man-in-the middle Attacks
- unauthorized VM Communication.

Study the following VM host utilization Snapshot and recommend Corrective action: CPU 92%, memory 88%, Storage I/O latency 35ms, average VM boot time 170S. Explain the likely bottlenecks

Ⓐ Introduction:-

In a Virtualized Enivornment, multiple virtual machines share the resources of a Physical host. Performance of monitorining helps identify bottlenecks that affects Systems Efficiency. The given metrics indicate that the host is under heavy load may Experience Performance issues.

Given Metrics:-

CPU utilization = 92%

Memory utilization = 88%

Storage I/O Latency = 35ms

Average VM Boot time = 170 Seconds

1. CPU Bottleneck:-

- CPU usage is Very high (92%)
- Processor is busy most of the time.
- Applications may run slowly.

2. Memory Bottleneck:-

- Memory usage is high (88%)
- less RAM is avaliable for new task.
- System Performance may decreases.

3. Storage Bottleneck:

- Storage latency is 35ms, which is high.
- Data reading and writing become slow.
- VM Startup takes more time

4. Slow VM Boot time:

- VM takes 170 seconds to start.
- Indicate CPU, memory, and storage Problems

Corrective Actions:

1. Increase CPU resources:
   - Add more CPU cores.
   - Move some VMs to another host.

2. Increase RAM:
   - Add more memory
   - Remove unused VMs.

3. Improve Storage:
   - use SSD & NVMe drives.
   - upgrade Storage Systems.

4. load balancing:
   - Use SSD (or) VMe across multiple servers.

5. Continuous Monitoring:

Regularly monitor CPU, memory, and storage usage.

Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with Compute, Storage, and network Virtualization considerations.

## Introduction:-

A Software defined Datacenter is a modern data Center where Compute, Storage, and networking resources are managed through software. unlike traditional dataCenters, most Operations are automated and centrally Controlled

## How SDDC improves Agility:-

1. faster resources Provisioning:-
   - New Virtual machines can be Created in min
   - No need to install new Physical hardware.

2. Easy Scalability:-
   - Resources Can be increased (or) decreased quickly
   - Supports changing workload demands.

3. Automation:-
   - Many tasks are Performed automatically.
   - Reduces manual efforts and errors.

4. Better resource utilization:

- Resources are Shared Efficiently.
- Reduces wastage of hardware.

Advantages of SDDC:

- faster Operations
- Reduced Costs.
- Better flexibility
- Improved Scalability
- Simplified management
- Higher Efficiency.

Conclusion:

A Software defined Datacenter (SDDC) improves agility by automating and Virtualizing Compute, storage, and networking resources Compared to a traditional datacenter. it provides faster deployment, easier Scalability, better Source utilization, and Simplified management, making it ideal for modern cloud Environments

Q) Multi-cloud federation Project fails due to identify mismatches between Providers. Analyze the issue and propose a secure identify and privacy design using federation concepts.

A **Introduction:-**

In a multi-cloud environment, service from different cloud Providers work together. Sometimes, users may have different identies (or) login methods con different cloud platforms. This creates identify mismatches and access problems.

**federated identify Solutions:-**

A federated identify managnment (FIM) system can slove this problem.

**How it works:-**

- users log in once using a central identify Provider
- The Identify provider verifies the user's identify
- Cloud Providers trust the identify Provider.
- user can access multiple cloud services without creating seperate accounts.

Components:-

1. Identify Provider (idp):- Authenticates users
2. Service Provider (sp):- Provides cloud Services
3. single sign-on (sso):- one login for multiple servics
4. Authentication Token:- Securelyshare identity informatin

## Advantages:-

- Single login for multiple clouds
- Better user Experience.
- Reduced Password management
- Easier administration.
- Improved interoperability between cloud providers.

## Privacy Implications:-

### Benefits:-

- Reduces duplicate user information.
- Better Control over user identities.
- Consistent Privacy Policies

### Challenges:-

- Personal information may be shared across Providers.
- Data leakage risks if identify data is not protected.
- Privacy regulation must be followed.

## Security and Privacy Measures:-

- use Multi-factor Authentication (MFA)
- Encrypt identify data and tokens
- Apply Role-based Access Control (RBAC).
- Regular Security audits
- Share only neccessary user information